

Untangling Protein Disorder Prediction

1 Background

Intrinsically disordered proteins (IDPs) and intrinsically disordered regions (IDRs) lack a fixed three-dimensional structure under physiological conditions, yet they play central roles in signaling, regulation, and disease. Predicting which residues are disordered has become a key task in structural bioinformatics, with dozens of predictors developed and regularly assessed in the community-wide CAID benchmarks¹.

A fundamental challenge is that “disorder” is not a single, well-defined property. Different experimental techniques capture different aspects of it, and therefore different datasets disagree on what counts as disordered. X-ray crystallography identifies *missing electron density* — regions too flexible to resolve. Curated databases like DisProt² aggregate evidence from diverse biophysical methods (NMR, CD, SAXS, proteolysis) into manually curated binary labels. AlphaFold2’s per-residue confidence score (pLDDT) has become a popular disorder proxy, but reflects prediction uncertainty rather than a direct measurement. Each source captures a different facet of the same biology, leading to inconsistent ground-truth annotations across datasets.

Among these techniques, NMR chemical shifts stand out because they provide a *continuous*, per-residue measure of disorder grounded in direct physical measurement. The CheZOD Z-score³ quantifies how much the experimentally observed backbone chemical shifts deviate from predicted random coil values⁴: small deviations indicate disorder ($Z < 3$), large deviations indicate stable structure ($Z > 8$), and values in between capture partial disorder. The original CheZOD database of 117 proteins was used to benchmark 26 disorder predictors⁵ and to train continuous-score predictors such as ODiNPred⁶ and SETH⁷.

TriZOD⁸ expands the CheZOD approach to the full Biological Magnetic Resonance Bank (BMRB;

¹Mehdiabadi, M., Del Conte, A., et al. “Critical Assessment of Protein Intrinsic Disorder Round 3.” *Proteins* 94(1), 414–424 (2026). <https://doi.org/10.1002/prot.70045>.

²Nugnes, M.V., Bouhraoua, K.E.A., et al. “DisProt in 2026: Enhancing Intrinsically Disordered Proteins Accessibility, Deposition, and Annotation.” *Nucleic Acids Research* 54(D1), D383–D392 (2026). <https://doi.org/10.1093/nar/gkaf1175>.

³Nielsen, J.T. & Mulder, F.A.A. “There Is Diversity in Disorder — ‘In All Chaos There Is a Cosmos, in All Disorder a Secret Order’.” *Frontiers in Molecular Biosciences* 3, 4 (2016). <https://doi.org/10.3389/fmolb.2016.00004>.

⁴Nielsen, J.T. & Mulder, F.A.A. “POTENCI: Prediction of Temperature, Neighbor and pH-Corrected Chemical Shifts for Intrinsically Disordered Proteins.” *J Biomol NMR* 70, 141–165 (2018). <https://doi.org/10.1007/s10858-018-0175-y>.

⁵Nielsen, J.T. & Mulder, F.A.A. “Quality and Bias of Protein Disorder Predictors.” *Scientific Reports* 9, 5137 (2019). <https://doi.org/10.1038/s41598-019-41644-w>.

⁶Dass, R., Mulder, F.A.A. & Nielsen, J.T. “ODiNPred: Comprehensive Prediction of Protein Order and Disorder.” *Scientific Reports* 10, 14780 (2020). <https://doi.org/10.1038/s41598-020-71716-1>.

⁷Ilzhoefer, D., Heinzinger, M. & Rost, B. “SETH Predicts Nuances of Residue Disorder from Protein Embeddings.” *Frontiers in Bioinformatics* 2, 1019597 (2022). <https://doi.org/10.3389/fbinf.2022.1019597>.

⁸Haak, M., Senoner, T., Schlensok, M., Wagemann, D. & Rost, B. “TriZOD Dataset.” *Figshare* (2024). <https://doi.org/10.6084/m9.figshare.25792035>. Code: <https://github.com/MarkusHaak/trizod>.

>17,800 entries), providing disorder scores for roughly ten times as many proteins. Its main contribution is the **G-score**, a new disorder metric computed from the same underlying secondary chemical shifts as the Z-score but whose expected value is independent of the number of measured backbone shifts, avoiding a systematic bias of the Z-score. With its [0, 1] range, it is better suited as a training target for machine learning. Four nested filtering tiers (unfiltered, tolerant, moderate, strict) let users trade dataset size for data quality. An updated version of TriZOD is currently under development: it integrates chemical shift re-referencing via LACS⁹ to correct the ~20% of ¹³C and ~35% of ¹⁵N BMRB entries with systematic referencing errors, along with improved filtering (e.g., excluding paramagnetic samples, correcting over-aggressive blacklists that previously removed valid entries).

UdonPred¹⁰ uses a deliberately simple architecture (a lightweight CNN on ProstT5 embeddings) to show that training data — not model capacity — is the main bottleneck in disorder prediction. Trained and cross-evaluated on seven datasets with divergent disorder definitions (TriZOD, CheZOD, SoftDis, PDBFlex, Atlas, pLDDT, DisProt), its 7×7 cross-dataset evaluation matrix demonstrates that a model trained on one definition of disorder often fails to generalize to another.

In this project, we will build on UdonPred and the updated TriZOD dataset to investigate how well current disorder predictors really perform, whether combining them helps, how much they agree, and whether their disagreement can be used to guide future experiments.

2 Datasets

- **UdonPred's 7 evaluation datasets:** TriZOD, CheZOD (original 117 proteins), SoftDis, PDBFlex, Atlas, pLDDT (from AlphaFold2), and DisProt. Code: <https://github.com/davidwagemann/udonpred>
- **TriZOD** (updated version): Per-residue G-scores (continuous disorder scores normalized to [0, 1]) recomputed from the full BMRB (~17,800 entries) with four filtering tiers. The updated pipeline integrates LACS re-referencing and improved sample filtering. Code: <https://github.com/MarkusHaak/trizod>; Dataset: <https://doi.org/10.6084/m9.figshare.25792035>
- **Human proteome:** The reviewed human proteome from UniProt (~20,000 proteins) for proteome-scale analysis.
- **CAID predictors:** For comparison, use top-performing methods from the latest CAID round¹.

3 Exploration Steps

1. Literature

- Familiarize yourself with the UdonPred preprint and its key references (Nielsen & Mulder 2016, 2019; ODINPred; SETH; the latest CAID round).

⁹Wang, L., Eghbalnia, H., Bahrami, A. & Markley, J.L. "Linear Analysis of Chemical Shift Differences and Its Application to the Detection and Correction of Errors in Referencing." *J Biomol NMR* 32, 13–22 (2005). <https://doi.org/10.1007/s10858-005-1717-0>.

¹⁰Schlensok, M., Wagemann, D., Senoner, T., Haak, M. & Rost, B. "UdonPred: Intrinsic Disorder Prediction — It's (Not About) the Architecture." *bioRxiv* (2026). <https://doi.org/10.64898/2026.01.26.701679>.



- Understand the TriZOD pipeline: how NMR chemical shifts become disorder scores (observed shifts → POTENCI random coil prediction → weighted secondary chemical shifts → offset correction → Z-score and G-score).
- Understand the evaluation protocol: what metrics are appropriate for continuous scores (Spearman correlation) vs. binary labels (AUROC, F1)?

2. Setup

- Set up the UdonPred training pipeline and reproduce its cross-dataset evaluation matrix (Figure 2).
- Generate the updated TriZOD dataset using the improved pipeline (with LACS re-referencing and new filters). Compare the resulting G-score distributions to the previous version to evaluate the impact of the improvements.
- Where duplicate BMRB entries exist (same protein measured independently), compare their G-scores to establish an experimental reproducibility baseline.

3. Establish baselines

Before evaluating more complex approaches, calibrate how much of UdonPred's performance is "real" vs. trivially achievable.

- Retrain UdonPred on each of the 7 datasets with **shuffled labels** to obtain an empirical null performance distribution.
- Implement simple baselines (e.g., amino acid composition + logistic regression, per-residue coil probability from a secondary structure predictor).
- Estimate an **experimental ceiling**: where the same protein has annotations in two datasets (e.g., DisProt $\cap$ TriZOD), compute the inter-annotation agreement as an upper bound on achievable performance.
- Re-express UdonPred's 7×7 matrix as *headroom above trivial baselines* rather than raw metric values.

4. Research Question: Can ensembles outperform individual models?

UdonPred trains seven separate models, one per dataset. Additionally, several strong disorder predictors are publicly available (see the CAID rankings¹). Can combining multiple predictions improve over any single model?

- Collect predictions from UdonPred's per-dataset models and from top-performing CAID methods on all 7 test sets.
- Compare ensembling strategies (e.g., simple averaging, learned weighting, stacking) and evaluate whether a single ensemble can generalize across all test sets.
- Assess whether adding external predictors improves over ensembling UdonPred models alone.

5. Research Question: Do predictors agree more than annotations do?

Different disorder datasets disagree because the underlying experiments measure different things. Do the *predictors* trained on these datasets also disagree — or have they learned a shared signal that transcends any single annotation source?

- Run UdonPred's per-dataset models and selected external predictors on the human proteome.



- Compute pairwise agreement between predictors (residue-level and protein-level) and compare it to agreement between the underlying experimental annotations.
- Identify “contested” regions where predictors strongly disagree. What characterizes these regions?

6. **Research Question: Can prediction disagreement guide new NMR experiments?**

NMR experiments are expensive. If predictors disagree about whether a region is disordered, that region is where a new measurement would be most informative.

- Design a scoring function that ranks protein fragments by expected information gain for disorder annotation — considering, e.g., predictor disagreement, predicted disorder near the decision boundary, fragment suitability for NMR (length, predicted solubility).
- Validate retrospectively: do the top-ranked fragments overlap with regions that later received BMRB chemical shift measurements?

4 Notes

- CheZOD Z-scores and TriZOD G-scores are continuous. When comparing to binary annotations (e.g., DisProt), standard thresholds are $Z = 8$ for Z-scores or $G = 0.5$ for G-scores, but the choice of threshold itself is worth investigating.
- When showing a specific protein from the dataset, indicate whether it was cherry-picked (and by which criteria) or chosen randomly.

